Assignment 2 Group 10 (Renk, Leon and Payal)

```
In [1]: import tarfile
        import pandas as pd
        import os
```

```
In [2]: def count_png_files_in_folder(folder_path):
            png_count = 0

            for root, dirs, files in os.walk(folder_path):
                png_count += sum(1 for file in files if file.endswith('.png'))
            return png_count

        count_png_files_in_folder("/Users/renkasik/Desktop/Intro To Focus Areas/data/BreaKHis_v1")
Out[2]: 7909
```

Import data to df with Benign/Malignant IDs and choosing magnification

```
In [3]: def extract_image_labels(folder_path):
            file_paths = []
            labels = []

            file_paths = []
            labels = []

            for root, dirs, files in os.walk(folder_path):
                for file_name in files:
                    if file_name.endswith('.png'):
                        file_path = os.path.join(root, file_name)
                        # Extract parts of the filename
                        parts = file_name.split('-')

                        label = parts[0].split('_')[1]  # 'M' or 'B' indicating malignancy
                        magnification = parts[3]  # Magnification levels

                        # Filter for images with 100x magnification
                        if magnification == '100':
                            labels.append(label)
                            file_paths.append(file_path)

            # Create a DataFrame with the extracted data
            df = pd.DataFrame({
                'FilePath': file_paths,
                'Label': labels
            })

            return df
```

```
In [4]: image_labels_df = extract_image_labels("/Users/renkasik/Desktop/Intro To Focus Areas/data/BreaKHis_v1")
```

```
In [5]: image_labels_df.head(5)
```

Out[5]:

	FilePath	Label
0	/Users/renkasik/Desktop/Intro To Focus Areas/d...	M
1	/Users/renkasik/Desktop/Intro To Focus Areas/d...	M
2	/Users/renkasik/Desktop/Intro To Focus Areas/d...	M
3	/Users/renkasik/Desktop/Intro To Focus Areas/d...	M
4	/Users/renkasik/Desktop/Intro To Focus Areas/d...	M

Split data into train. test and validation sets

```
In [6]: from sklearn.model_selection import train_test_split
```

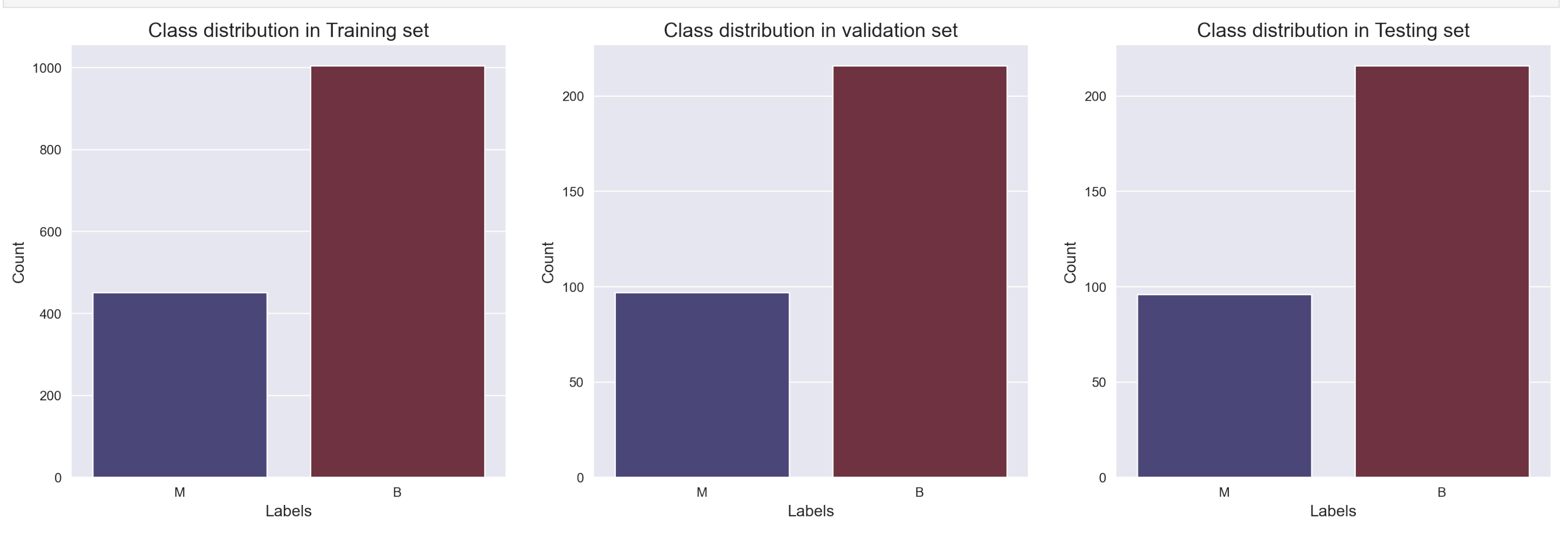
```
In [7]: labels = image_labels_df['Label']

        X_train, X_temp = train_test_split(image_labels_df, test_size=0.3, stratify=labels, random_state = 42)

        label_test_val = X_temp['Label']

        X_test, X_val = train_test_split(X_temp, test_size=0.5, stratify=label_test_val, random_state = 42)

        print('The shape of train data',X_train.shape)
        print('The shape of test data',X_test.shape)
        print('The shape of validation data',X_val.shape)

        The shape of train data (1456, 2)
        The shape of test data (312, 2)
        The shape of validation data (313, 2)
```

```
In [8]: # Convert Labels to Numeric
        label_mapping = {'M': 1, 'B': 0}
        X_train['Label'] = X_train['Label'].map(label_mapping)
        X_val['Label'] = X_val['Label'].map(label_mapping)
        X_test['Label'] = X_test['Label'].map(label_mapping)
```

Distribution of train, test and validation Datasets

```
In [9]: import numpy as np
        import matplotlib.pyplot as plt
        import seaborn as sns
```

```
In [10]: labels = ['M','B']

         #count unique label appearances
         label1,count1 = np.unique(X_train.Label,return_counts=True)
         label2,count2 = np.unique(X_val.Label,return_counts=True)
         label3,count3 = np.unique(X_test.Label,return_counts=True)

         # DataFrames with 1 column for count of labels, label names in index
         uni1 = pd.DataFrame(data=count1,index=labels,columns=['Count1'])
         uni2 = pd.DataFrame(data=count2,index=labels,columns=['Count2'])
         uni3 = pd.DataFrame(data=count3,index=labels,columns=['Count3'])

         plt.figure(figsize=(20,6),dpi=200)
         sns.set_style('darkgrid')

         plt.subplot(131)
         sns.barplot(data=uni1,x=uni1.index, y='Count1', hue=uni1.index, palette='icefire', legend=False).set_title('Class distribution in Training set',fontsize=15)
         plt.xlabel('Labels',fontsize=12)
         plt.ylabel('Count',fontsize=12)

         plt.subplot(132)
         sns.barplot(data=uni2,x=uni2.index, y='Count2', hue=uni1.index, palette='icefire', legend=False).set_title('Class distribution in validation set',fontsize=15)
         plt.xlabel('Labels',fontsize=12)
         plt.ylabel('Count',fontsize=12)


         plt.subplot(133)
         sns.barplot(data=uni3,x=uni3.index, y='Count3', hue=uni1.index, palette='icefire', legend=False).set_title('Class distribution in Testing set',fontsize=15)
         plt.xlabel('Labels',fontsize=12)
         plt.ylabel('Count',fontsize=12)

         plt.show()
```



Deep Learning based Classifiers

```
In [11]: import tensorflow as tf
         from tensorflow.keras import datasets, layers, models
         from tensorflow.keras.metrics import Precision, Recall
```

CNN

```
In [12]: # Load and preprocess data
         IMG_SIZE = (32, 32)  # Adjust

         def load_and_preprocess_image(filepath):
             image = tf.io.read_file(filepath)
             image = tf.image.decode_png(image, channels=3)
             image = tf.image.resize(image, IMG_SIZE)
             image = image / 255.0 # Normalize pixel values to [0, 1]
             return image
```

```
In [13]: base_dir = "/Users/renkasik/Desktop/Intro To Focus Areas/data/BreaKHis_v1"
```

```
In [14]: # Convert DataFrame columns to lists
         train_filepaths = X_train['FilePath'].tolist()
         train_labels = X_train['Label'].tolist()

         val_filepaths = X_val['FilePath'].tolist()
         val_labels = X_val['Label'].tolist()

         test_filepaths = X_test['FilePath'].tolist()
         test_labels = X_test['Label'].tolist()

         # load images and labels as TensorFlow Datasets
         def create_dataset(filepaths, labels):
             # Create TensorFlow datasets from paths and labels
             filepaths_ds = tf.data.Dataset.from_tensor_slices(filepaths)
             images_ds = filepaths_ds.map(load_and_preprocess_image, num_parallel_calls=tf.data.experimental.AUTOTUNE)
             labels_ds = tf.data.Dataset.from_tensor_slices(labels)

             # Zip the datasets together
             return tf.data.Dataset.zip((images_ds, labels_ds))
```

```
In [15]: i = 0
         for filepath in train_filepaths:
             if not os.path.exists(filepath):
                 print(f"File not found: {filepath}")
```

```
In [16]: BATCH_SIZE = 32  # Set the batch size according to memory capacity

         # Shuffle, batch, and prefetch the datasets
         train_ds = create_dataset(train_filepaths, train_labels).shuffle(buffer_size=len(train_filepaths)).batch(BATCH_SIZE).prefetch(tf.data.experimental.AUTOTUNE)
         val_ds = create_dataset(val_filepaths, val_labels).batch(BATCH_SIZE).prefetch(tf.data.experimental.AUTOTUNE)
         test_ds = create_dataset(test_filepaths, test_labels).batch(BATCH_SIZE).prefetch(tf.data.experimental.AUTOTUNE)

         2024-11-03 22:46:26.680630: I tensorflow/core/platform/cpu_feature_guard.cc:193] This TensorFlow binary is optimized with oneAPI Deep Neural Network Library (oneDNN) to use the
         following CPU instructions in performance-critical operations:  SSE4.1 SSE4.2
         To enable them in other operations, rebuild TensorFlow with the appropriate compiler flags.
```

```
In [17]: model = models.Sequential([
             layers.Conv2D(32, (3, 3), activation='relu', input_shape=(32, 32, 3)),  # Adjust input_shape to image dimensions, (3,3) is filter size, 3 is RGB color channels
             layers.MaxPooling2D((2, 2)), # reduces each dimension (height and width) by half, summarizing the main features
             layers.Conv2D(64, (3, 3), activation='relu'), # 64 filters
             layers.MaxPooling2D((2, 2)),
             layers.Conv2D(64, (3, 3), activation='relu'),
             layers.Flatten(),
             layers.Dense(64, activation='relu'), # number of neurons in the layer
             layers.Dense(1, activation='sigmoid')
         ])
```

```
In [18]: model.compile(optimizer='adam',
                       loss='binary_crossentropy',
                       metrics=['accuracy', Precision(), Recall()])
```

```
In [19]: history = model.fit(train_ds,
                             validation_data=val_ds,
                             epochs=10)

         Epoch 1/10
         46/46 [==============================] - 4s 24ms/step - loss: 0.6124 - accuracy: 0.6820 - precision: 0.6903 - recall: 0.9781 - val_loss: 0.5315 - val_accuracy: 0.6901 - val_prec
         ision: 0.6901 - val_recall: 1.0000
         Epoch 2/10
         46/46 [==============================] - 3s 19ms/step - loss: 0.5503 - accuracy: 0.6909 - precision: 0.7170 - recall: 0.9124 - val_loss: 0.4987 - val_accuracy: 0.7061 - val_prec
         ision: 0.7013 - val_recall: 1.0000
         Epoch 3/10
         46/46 [==============================] - 3s 19ms/step - loss: 0.5049 - accuracy: 0.7706 - precision: 0.7846 - recall: 0.9204 - val_loss: 0.4445 - val_accuracy: 0.8147 - val_prec
         ision: 0.8559 - val_recall: 0.8796
         Epoch 4/10
         46/46 [==============================] - 3s 21ms/step - loss: 0.4855 - accuracy: 0.8001 - precision: 0.8121 - recall: 0.9244 - val_loss: 0.4489 - val_accuracy: 0.8019 - val_prec
         ision: 0.7873 - val_recall: 0.9769
         Epoch 5/10
         46/46 [==============================] - 3s 20ms/step - loss: 0.4713 - accuracy: 0.7919 - precision: 0.8005 - recall: 0.9303 - val_loss: 0.4312 - val_accuracy: 0.8211 - val_prec
         ision: 0.8333 - val_recall: 0.9259
         Epoch 6/10
         46/46 [==============================] - 3s 20ms/step - loss: 0.4534 - accuracy: 0.8132 - precision: 0.8258 - recall: 0.9244 - val_loss: 0.4296 - val_accuracy: 0.8147 - val_prec
         ision: 0.8347 - val_recall: 0.9120
         Epoch 7/10
         46/46 [==============================] - 3s 20ms/step - loss: 0.4608 - accuracy: 0.8063 - precision: 0.8254 - recall: 0.9124 - val_loss: 0.4814 - val_accuracy: 0.7955 - val_prec
         ision: 0.8585 - val_recall: 0.8426
         Epoch 8/10
         46/46 [==============================] - 3s 20ms/step - loss: 0.4585 - accuracy: 0.7995 - precision: 0.8197 - recall: 0.9095 - val_loss: 0.4436 - val_accuracy: 0.8179 - val_prec
         ision: 0.8502 - val_recall: 0.8935
         Epoch 9/10
         46/46 [==============================] - 3s 19ms/step - loss: 0.4548 - accuracy: 0.8111 - precision: 0.8247 - recall: 0.9224 - val_loss: 0.4214 - val_accuracy: 0.8179 - val_prec
         ision: 0.8472 - val_recall: 0.8981
         Epoch 10/10
         46/46 [==============================] - 3s 20ms/step - loss: 0.4738 - accuracy: 0.7946 - precision: 0.8209 - recall: 0.8985 - val_loss: 0.4576 - val_accuracy: 0.7923 - val_prec
         ision: 0.7745 - val_recall: 0.9861
```

```
In [20]: results = model.evaluate(test_ds)
         metrics = ['Loss', 'Accuracy', 'Precision', 'Recall(Sensitivity)']
         for i in range(len(results)):
             print(metrics[i] + ': ' + str(results[i]))

         10/10 [==============================] - 1s 47ms/step - loss: 0.4197 - accuracy: 0.8301 - precision: 0.8123 - recall: 0.9815
         Loss: 0.4197402894496918
         Accuracy: 0.8301281929016113
         Precision: 0.8122605085372925
         Recall(Sensitivity): 0.9814814925193787
```

Fully connected NN with one hidden layer

```
In [21]: model_2 = models.Sequential([
             layers.Flatten(input_shape=(32, 32, 3)),  # fully connected -> flatten from 2D to 1D
             layers.Dense(64, activation='relu'),  # hidden layer -> learning features from the input data
             layers.Dense(1, activation='sigmoid')
         ])
```

```
In [22]: model_2.compile(optimizer='adam',
                         loss='binary_crossentropy',
                         metrics=['accuracy', Precision(), Recall()])
```

```
In [27]: history_2 = model_2.fit(train_ds,
                             validation_data=val_ds,
                             epochs=10)

         Epoch 1/10
         46/46 [==============================] - 3s 11ms/step - loss: 0.6636 - accuracy: 0.6882 - precision_1: 0.7181 - recall_1: 0.9025 - val_loss: 0.5449 - val_accuracy: 0.6901 - val_
         precision_1: 0.6901 - val_recall_1: 1.0000
         Epoch 2/10
         46/46 [==============================] - 2s 11ms/step - loss: 0.5282 - accuracy: 0.7521 - precision_1: 0.7683 - recall_1: 0.9174 - val_loss: 0.5204 - val_accuracy: 0.6965 - val_
         precision_1: 0.6945 - val_recall_1: 1.0000
         Epoch 3/10
         46/46 [==============================] - 2s 11ms/step - loss: 0.5315 - accuracy: 0.7376 - precision_1: 0.7754 - recall_1: 0.8726 - val_loss: 0.5490 - val_accuracy: 0.6901 - val_
         precision_1: 0.6901 - val_recall_1: 1.0000
         Epoch 4/10
         46/46 [==============================] - 2s 11ms/step - loss: 0.5168 - accuracy: 0.7754 - precision_1: 0.7995 - recall_1: 0.9005 - val_loss: 0.5154 - val_accuracy: 0.7157 - val_
         precision_1: 0.7096 - val_recall_1: 0.9954
         Epoch 5/10
         46/46 [==============================] - 2s 11ms/step - loss: 0.5029 - accuracy: 0.7665 - precision_1: 0.7919 - recall_1: 0.8975 - val_loss: 0.4755 - val_accuracy: 0.7891 - val_
         precision_1: 0.7930 - val_recall_1: 0.9398
         Epoch 6/10
         46/46 [==============================] - 2s 11ms/step - loss: 0.4836 - accuracy: 0.7946 - precision_1: 0.8124 - recall_1: 0.9134 - val_loss: 0.4803 - val_accuracy: 0.7923 - val_
         precision_1: 0.7786 - val_recall_1: 0.9769
         Epoch 7/10
         46/46 [==============================] - 2s 12ms/step - loss: 0.5042 - accuracy: 0.7713 - precision_1: 0.7963 - recall_1: 0.8985 - val_loss: 0.7030 - val_accuracy: 0.6198 - val_
         precision_1: 0.9292 - val_recall_1: 0.4861
         Epoch 8/10
         46/46 [==============================] - 2s 11ms/step - loss: 0.5257 - accuracy: 0.7438 - precision_1: 0.7915 - recall_1: 0.8537 - val_loss: 0.4940 - val_accuracy: 0.7764 - val_
         precision_1: 0.7589 - val_recall_1: 0.9907
         Epoch 9/10
         46/46 [==============================] - 2s 11ms/step - loss: 0.4995 - accuracy: 0.7734 - precision_1: 0.7984 - recall_1: 0.8985 - val_loss: 0.6085 - val_accuracy: 0.6486 - val_
         precision_1: 0.9077 - val_recall_1: 0.5463
         Epoch 10/10
         46/46 [==============================] - 2s 11ms/step - loss: 0.5183 - accuracy: 0.7486 - precision_1: 0.7939 - recall_1: 0.8587 - val_loss: 0.4783 - val_accuracy: 0.7891 - val_
         precision_1: 0.7820 - val_recall_1: 0.9630
```

```
In [28]: results_2 = model_2.evaluate(test_ds)
         metrics = ['Loss', 'Accuracy', 'Precision', 'Recall(Sensitivity)']
         for i in range(len(results_2)):
             print(metrics[i] + ': ' + str(results_2[i]))

         10/10 [==============================] - 0s 43ms/step - loss: 0.4393 - accuracy: 0.8429 - precision_1: 0.8465 - recall_1: 0.9444
         Loss: 0.4392950236797333
         Accuracy: 0.8429487347602844
         Precision: 0.8464730381965637
         Recall(Sensitivity): 0.9444444179534912
```

ResNet-50

```
In [25]: from tensorflow.keras.applications import ResNet50
```

```
In [34]: IMG_SIZE_resnet = (224, 224) #ResNet50 input size requirement

         def load_and_preprocess_image_resnet(filepath):
             image = tf.io.read_file(filepath)
             image = tf.image.decode_png(image, channels=3)
             image = tf.image.resize(image, IMG_SIZE_resnet)
             image = tf.keras.applications.resnet50.preprocess_input(image) #ResNet preprocessing
             return image
```

```
In [39]: def create_dataset_resnet(filepaths, labels):
             # Create TensorFlow datasets from paths and labels for ResNet
             filepaths_ds = tf.data.Dataset.from_tensor_slices(filepaths)
             images_ds = filepaths_ds.map(load_and_preprocess_image_resnet, num_parallel_calls=tf.data.experimental.AUTOTUNE)
             labels_ds = tf.data.Dataset.from_tensor_slices(labels)

             # Zip the datasets together
             return tf.data.Dataset.zip((images_ds, labels_ds))
```

```
In [40]: # Shuffle, batch, and prefetch the datasets
         train_ds_resnet = create_dataset_resnet(train_filepaths, train_labels).shuffle(buffer_size=len(train_filepaths)).batch(BATCH_SIZE).prefetch(tf.data.experimental.AUTOTUNE)
         val_ds_resnet = create_dataset_resnet(val_filepaths, val_labels).batch(BATCH_SIZE).prefetch(tf.data.experimental.AUTOTUNE)
         test_ds_resnet = create_dataset_resnet(test_filepaths, test_labels).batch(BATCH_SIZE).prefetch(tf.data.experimental.AUTOTUNE)
```

```
In [43]: base_model = ResNet50(weights='imagenet', include_top=False, input_shape=(224, 224, 3)) #without the top classification layer to add custom classification layers later
         base_model.trainable = False  #freeze base model to avoid fine tuning
```

```
In [44]: model_3 = models.Sequential([
             base_model,
             layers.GlobalAveragePooling2D(), #reduce output to single vector
             layers.Dense(1, activation='sigmoid')
         ])
```

```
In [45]: model_3.compile(optimizer='adam',
                         loss='binary_crossentropy',
                         metrics=['accuracy', Precision(), Recall()])
```

```
In [46]: history_3 = model_3.fit(train_ds_resnet,
                             validation_data=val_ds_resnet,
                             epochs=10)

         Epoch 1/10
         46/46 [==============================] - 84s 2s/step - loss: 0.4806 - accuracy: 0.7788 - precision_2: 0.7952 - recall_2: 0.9154 - val_loss: 0.3903 - val_accuracy: 0.8466 - val_p
         recision_2: 0.8750 - val_recall_2: 0.9074
         Epoch 2/10
         46/46 [==============================] - 81s 2s/step - loss: 0.3158 - accuracy: 0.8695 - precision_2: 0.8812 - recall_2: 0.9373 - val_loss: 0.3241 - val_accuracy: 0.8626 - val_p
         recision_2: 0.9061 - val_recall_2: 0.8935
         Epoch 3/10
         46/46 [==============================] - 83s 2s/step - loss: 0.2637 - accuracy: 0.9038 - precision_2: 0.9139 - recall_2: 0.9502 - val_loss: 0.2798 - val_accuracy: 0.9137 - val_p
         recision_2: 0.9163 - val_recall_2: 0.9630
         Epoch 4/10
         46/46 [==============================] - 101s 2s/step - loss: 0.2247 - accuracy: 0.9196 - precision_2: 0.9237 - recall_2: 0.9632 - val_loss: 0.2682 - val_accuracy: 0.9042 - val_
         precision_2: 0.9429 - val_recall_2: 0.9167
         Epoch 5/10
         46/46 [==============================] - 99s 2s/step - loss: 0.2050 - accuracy: 0.9265 - precision_2: 0.9309 - recall_2: 0.9652 - val_loss: 0.2489 - val_accuracy: 0.9201 - val_p
         recision_2: 0.9484 - val_recall_2: 0.9352
         Epoch 6/10
         46/46 [==============================] - 102s 2s/step - loss: 0.1830 - accuracy: 0.9423 - precision_2: 0.9484 - recall_2: 0.9692 - val_loss: 0.2314 - val_accuracy: 0.9361 - val_
         precision_2: 0.9537 - val_recall_2: 0.9537
         Epoch 7/10
         46/46 [==============================] - 98s 2s/step - loss: 0.1705 - accuracy: 0.9505 - precision_2: 0.9542 - recall_2: 0.9751 - val_loss: 0.2205 - val_accuracy: 0.9233 - val_p
         recision_2: 0.9211 - val_recall_2: 0.9722
         Epoch 8/10
         46/46 [==============================] - 99s 2s/step - loss: 0.1598 - accuracy: 0.9533 - precision_2: 0.9607 - recall_2: 0.9721 - val_loss: 0.2214 - val_accuracy: 0.9265 - val_p
         recision_2: 0.9573 - val_recall_2: 0.9352
         Epoch 9/10
         46/46 [==============================] - 99s 2s/step - loss: 0.1480 - accuracy: 0.9588 - precision_2: 0.9601 - recall_2: 0.9811 - val_loss: 0.2043 - val_accuracy: 0.9457 - val_p
         recision_2: 0.9543 - val_recall_2: 0.9676
         Epoch 10/10
         46/46 [==============================] - 96s 2s/step - loss: 0.1374 - accuracy: 0.9602 - precision_2: 0.9638 - recall_2: 0.9791 - val_loss: 0.2015 - val_accuracy: 0.9425 - val_p
         recision_2: 0.9541 - val_recall_2: 0.9630
```

```
In [48]: #fine tuning
         base_model.trainable = True
         model_3.compile(optimizer=tf.keras.optimizers.Adam(1e-5),  #lower learning rate
                         loss='binary_crossentropy',
                         metrics=['accuracy', Precision(), Recall()])

         history_3 = model_3.fit(train_ds_resnet,
                             validation_data=val_ds_resnet,
                             epochs=5)

         Epoch 1/5
         46/46 [==============================] - 288s 6s/step - loss: 0.5493 - accuracy: 0.7548 - precision_4: 0.9589 - recall_4: 0.6736 - val_loss: 0.3049 - val_accuracy: 0.8722 - val_
         precision_4: 0.8465 - val_recall_4: 0.9954
         Epoch 2/5
         46/46 [==============================] - 281s 6s/step - loss: 0.1458 - accuracy: 0.9602 - precision_4: 0.9876 - recall_4: 0.9542 - val_loss: 0.2671 - val_accuracy: 0.8882 - val_
         precision_4: 0.8664 - val_recall_4: 0.9907
         Epoch 3/5
         46/46 [==============================] - 285s 6s/step - loss: 0.0737 - accuracy: 0.9890 - precision_4: 0.9920 - recall_4: 0.9920 - val_loss: 0.2373 - val_accuracy: 0.9042 - val_
         precision_4: 0.8875 - val_recall_4: 0.9861
         Epoch 4/5
         46/46 [==============================] - 284s 6s/step - loss: 0.0433 - accuracy: 0.9973 - precision_4: 0.9970 - recall_4: 0.9990 - val_loss: 0.1919 - val_accuracy: 0.9201 - val_
         precision_4: 0.9064 - val_recall_4: 0.9861
         Epoch 5/5
         46/46 [==============================] - 285s 6s/step - loss: 0.0298 - accuracy: 0.9993 - precision_4: 0.9990 - recall_4: 1.0000 - val_loss: 0.1611 - val_accuracy: 0.9361 - val_
         precision_4: 0.9261 - val_recall_4: 0.9861
```

```
In [49]: results_3 = model_3.evaluate(test_ds_resnet)
         metrics = ['Loss', 'Accuracy', 'Precision', 'Recall(Sensitivity)']
         for i in range(len(results_3)):
             print(metrics[i] + ': ' + str(results_3[i]))

         10/10 [==============================] - 14s 1s/step - loss: 0.1974 - accuracy: 0.9199 - precision_4: 0.9099 - recall_4: 0.9815
         Loss: 0.19740530848503113
         Accuracy: 0.9198718070983887
         Precision: 0.9098712205886841
         Recall(Sensitivity): 0.9814814925193787
```

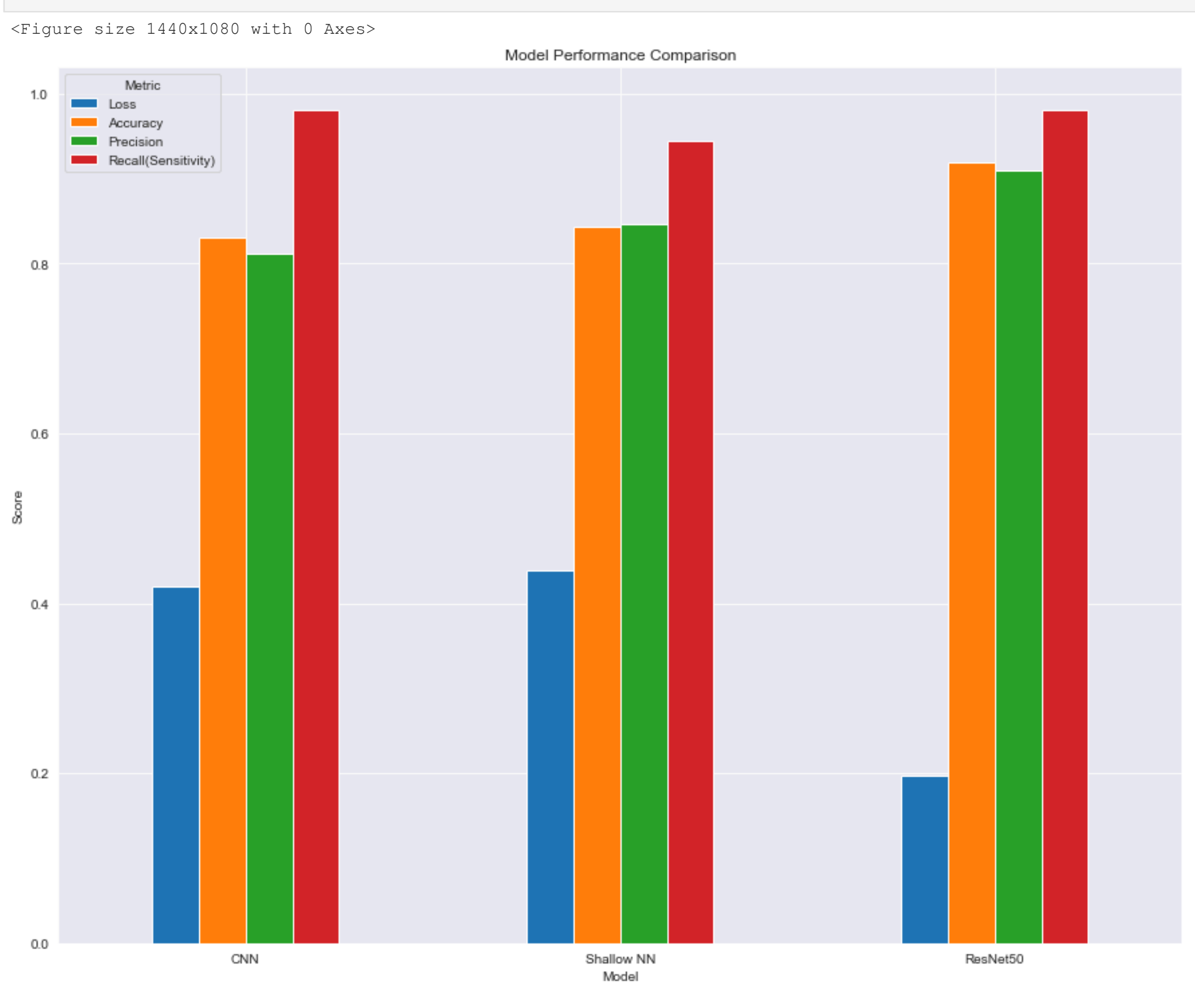
Plots

```
In [53]: results = np.array(results)
         results_2 = np.array(results_2)
         results_3 = np.array(results_3)
         results_df = pd.DataFrame({'CNN': results, 'Shallow NN': results_2, 'ResNet50': results_3})

         results_df = results_df.T

         results_df.columns = ['Loss', 'Accuracy', 'Precision', 'Recall(Sensitivity)']
         results_df
```

Out[53]:

	Loss	Accuracy	Precision	Recall(Sensitivity)
CNN	0.419740	0.830128	0.812261	0.981481
Shallow NN	0.439295	0.842949	0.846473	0.944444
ResNet50	0.197405	0.919872	0.909871	0.981481

```
In [61]: plt.figure(figsize=(20, 15))

         results_df.plot(kind='bar', rot=0, figsize=(15,12))
         plt.title('Model Performance Comparison')
         plt.ylabel('Score')
         plt.xlabel('Model')

         # Show the plot
         plt.legend(title='Metric')
         plt.show()

         <Figure size 1440x1080 with 0 Axes>
```



```
In [71]: from sklearn.metrics import roc_curve, auc
```

```
In [69]: X_test_images = np.array([load_and_preprocess_image(fp) for fp in X_test['FilePath']])
         X_test_images_resnet = np.array([load_and_preprocess_image_resnet(fp) for fp in X_test['FilePath']])

         probabilities_CNN = model.predict(X_test_images)
         probabilities_ShallowNN = model_2.predict(X_test_images)
         probabilities_ResNet50 = model_3.predict(X_test_images_resnet)

         10/10 [==============================] - 0s 5ms/step
         10/10 [==============================] - 0s 931us/step
         10/10 [==============================] - 15s 1s/step
```

```
In [73]: fpr_CNN, tpr_CNN, _ = roc_curve(test_labels, probabilities_CNN)
         fpr_ShallowNN, tpr_ShallowNN, _ = roc_curve(test_labels, probabilities_ShallowNN)
         fpr_ResNet50, tpr_ResNet50, _ = roc_curve(test_labels, probabilities_ResNet50)

         auc_CNN = auc(fpr_CNN, tpr_CNN)
         auc_ShallowNN = auc(fpr_ShallowNN, tpr_ShallowNN)
         auc_ResNet50 = auc(fpr_ResNet50, tpr_ResNet50)
```

```
In [74]: plt.figure(figsize=(10, 8))


         plt.plot(fpr_CNN, tpr_CNN, label=f'CNN (AUC = {auc_CNN:.2f})')
         plt.plot(fpr_ShallowNN, tpr_ShallowNN, label=f'Shallow NN (AUC = {auc_ShallowNN:.2f})')
         plt.plot(fpr_ResNet50, tpr_ResNet50, label=f'ResNet50 (AUC = {auc_ResNet50:.2f})')

         plt.plot([0, 1], [0, 1], 'k--', label='Random (AUC = 0.50)')


         plt.xlabel('False Positive Rate')
         plt.ylabel('True Positive Rate')
         plt.title('ROC Curve Comparison of Models')
         plt.legend(loc='lower right')
         plt.show()
```